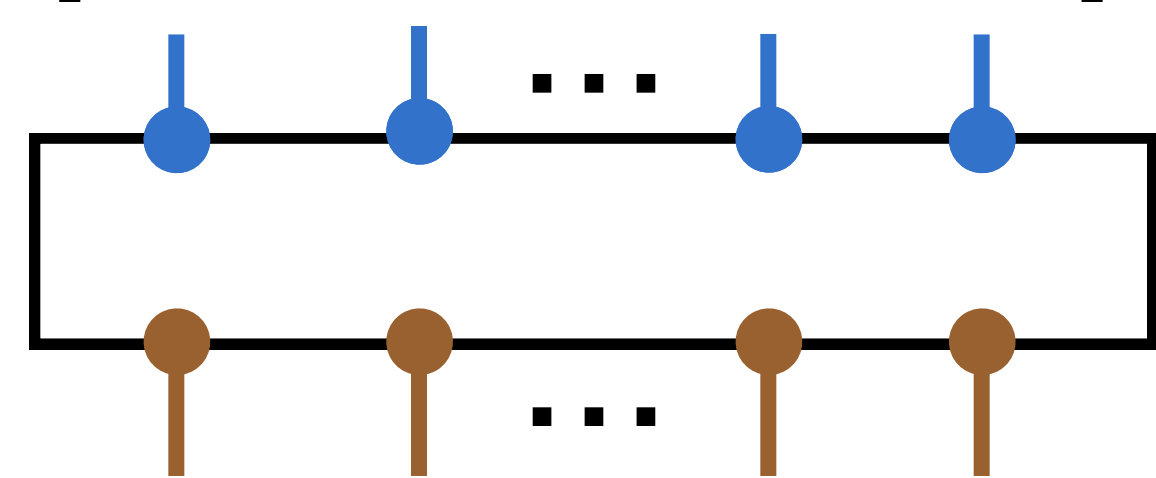
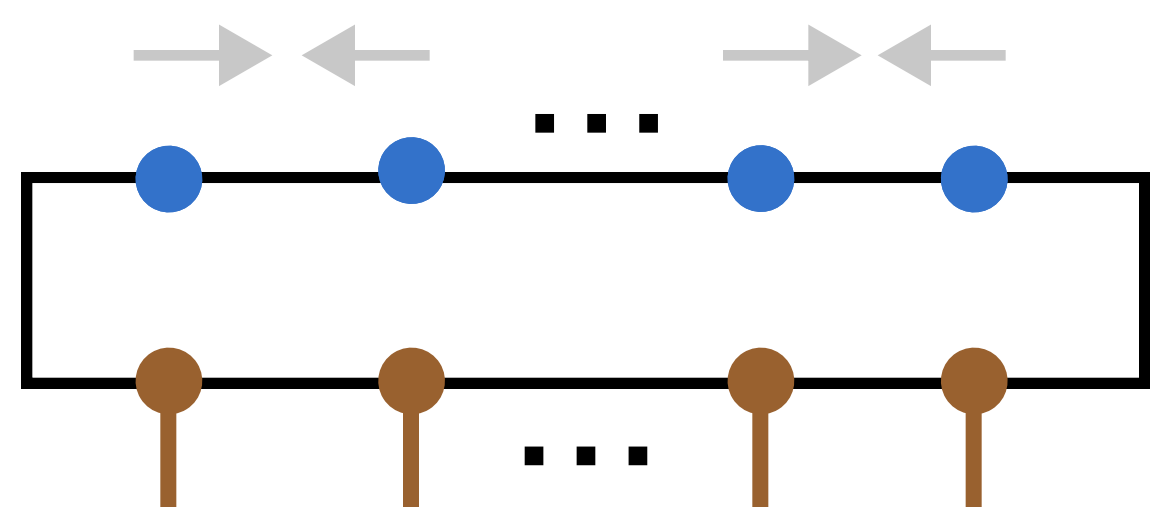


Input: syndrome $\vec{s}$

$$\vec{s} = [0, 1, \dots, 1, 0]^T$$

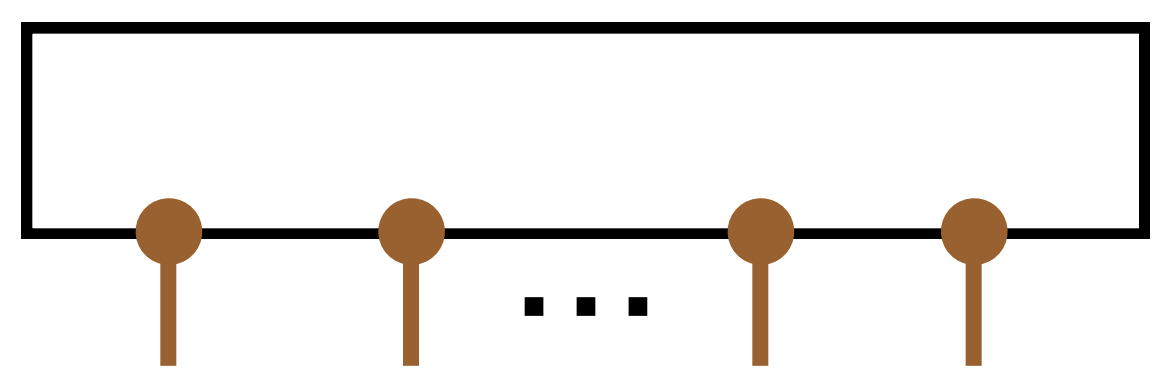


matrix product state



parallel contraction

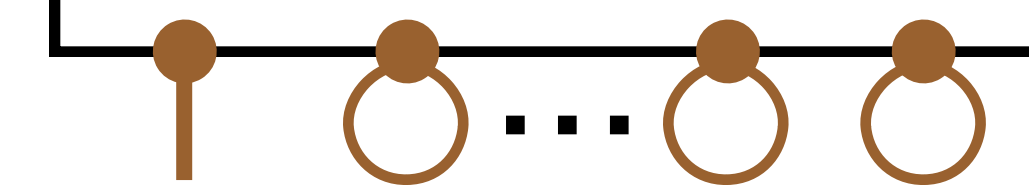
$$\downarrow O(M\chi^2)$$



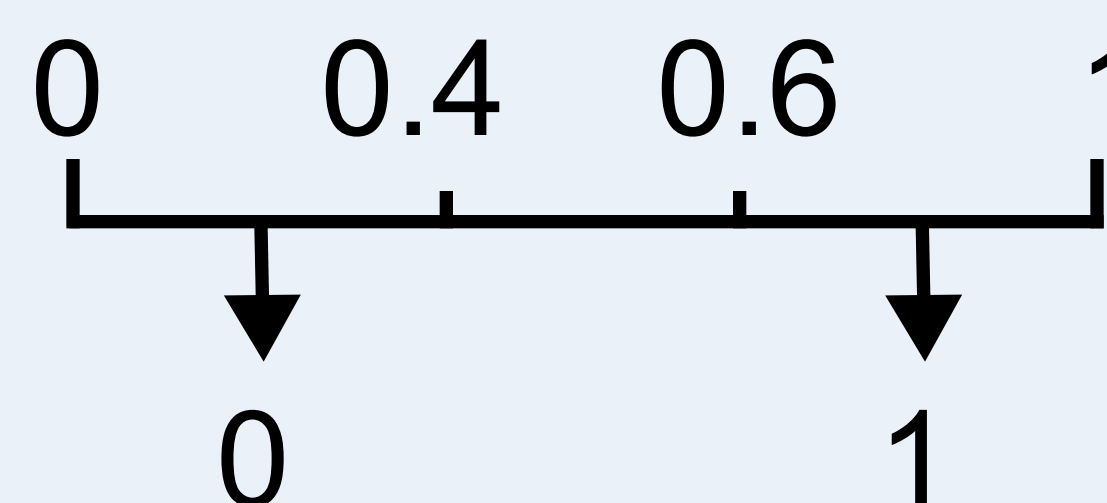
MPS with logical indices

trace other indices to get marginal probability

$$P(l_1 = 1 | \vec{s})$$

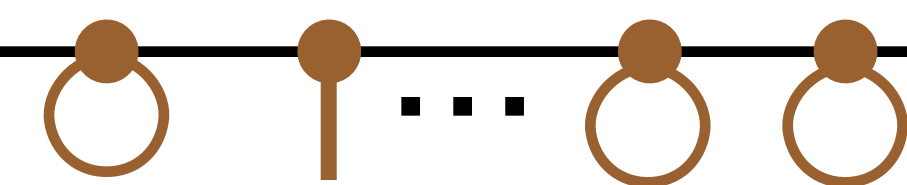


$$O(K\chi^4) \downarrow 0.78$$

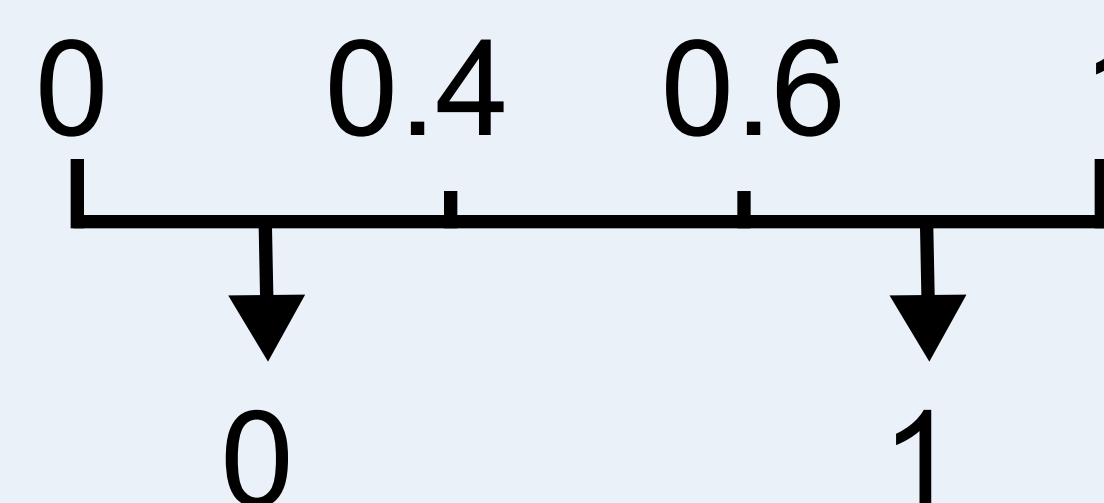


$$\downarrow l_1 = 1$$

$$P(l_2 = 1 | \vec{s})$$

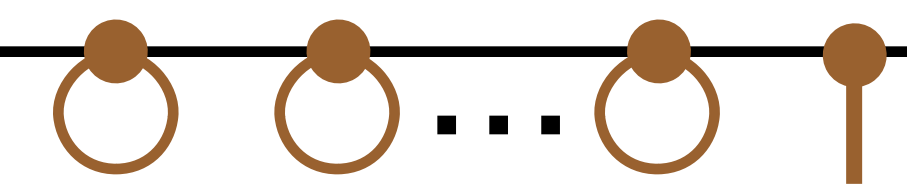


$$O(K\chi^4) \downarrow 0.52$$

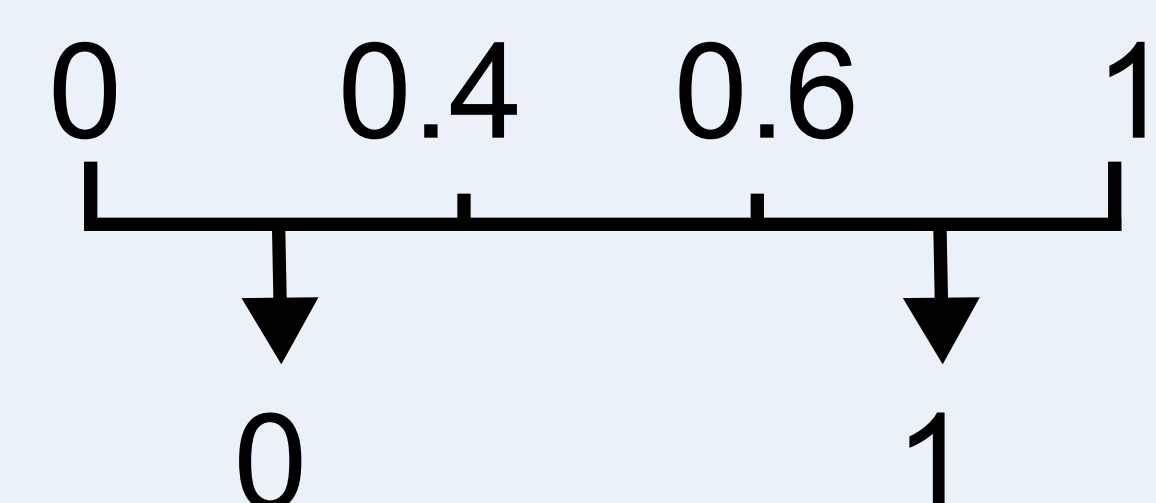


$$\downarrow l_2 \text{ unknown}$$

$$\dots P(l_K = 1 | \vec{s})$$



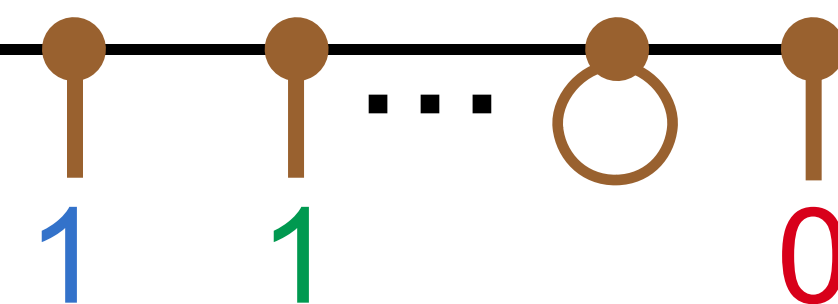
$$O(K\chi^4) \downarrow 0.12$$



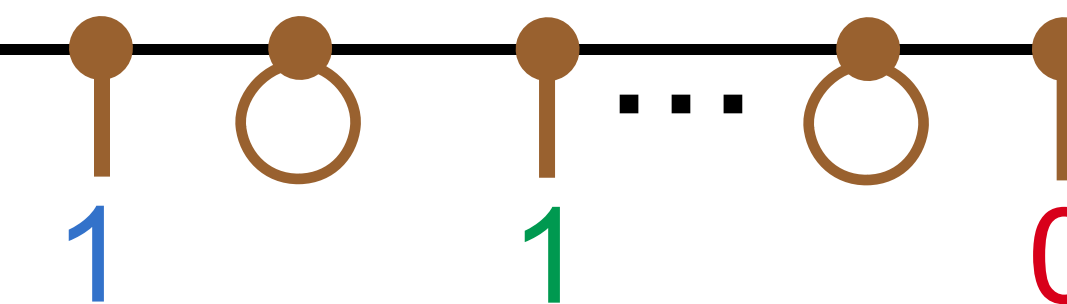
$$\downarrow l_K = 1$$

get conditional probability for unknown indices

$$P(l_2 = 1 | l_1 = 1, l_K = 0, \vec{s})$$



$$P(l_3 = 1 | l_1 = 1, l_K = 0, \vec{s})$$



repeat until all indices are known

Output: most likely logical operator $\vec{l}$